Topic-Wise Questions

EC2 (Elastic Compute Cloud)

1. **What are the different EC2 instance types and their use cases?**
   - *Hint: General Purpose (t3, m5), Compute Optimized (c5), Memory Optimized (r5, x1), Storage Optimized (i3, d2), Accelerated Computing (p3, g4)*
2. **Explain EC2 placement groups and their types.**
   - *Hint: Cluster (low latency), Partition (fault tolerance), Spread (critical instances)*
3. **What is the difference between EBS-backed and Instance Store-backed AMIs?**
   - *Hint: EBS = persistent, Instance Store = ephemeral*
4. **How would you implement zero-downtime patching for a fleet of 1000+ EC2 instances across multiple AZs?**
   - *Hint: Systems Manager Patch Manager + Auto Scaling Groups + rolling updates + health checks + blue-green deployment*
5. **Explain EC2 Dedicated Hosts vs Dedicated Instances and their compliance implications.**
   - *Hint: Dedicated Hosts = physical server control, licensing, compliance; Dedicated Instances = isolated hardware*
6. **How do you optimize EC2 costs for workloads with predictable usage patterns vs unpredictable spiky workloads?**
   - *Hint: Reserved Instances + Savings Plans for predictable; Spot Instances + Auto Scaling for spiky*
7. **Design a solution for EC2 instances to securely access S3 without storing credentials.**
   - *Hint: IAM roles, instance profiles, temporary credentials, least privilege*
8. **How would you troubleshoot an EC2 instance that's unreachable but shows as running in the console?**
   - *Hint: Security Groups, NACLs, OS-level firewall, system logs, instance status checks*

IAM (Identity and Access Management)

1. **What is the difference between IAM roles and IAM users?**
   - *Hint: Users = permanent identity, Roles = temporary credentials for services/cross-account*
2. **Explain IAM policy evaluation logic.**
   - *Hint: Explicit Deny > Explicit Allow > Default Deny*
3. **What are IAM policy conditions and give examples?**
   - *Hint: StringEquals, IpAddress, DateGreaterThan, MFA conditions*
4. **Design a complex IAM policy that allows developers to manage only their own resources tagged with their username.**
   - *Hint: Resource-based conditions, StringEquals with aws:RequestedRegion, aws:userid, dynamic policy variables*

5. **How would you implement cross-account access for a CI/CD pipeline that deploys to multiple AWS accounts?**
    - *Hint: Cross-account roles, external ID, assume role chains, temporary credentials*
6. **Explain IAM policy boundaries and how they differ from regular IAM policies.**
    - *Hint: Permission boundaries = maximum permissions, intersection with identity-based policies*
7. **How do you implement IAM policy versioning and rollback strategies for production environments?**
    - *Hint: Policy versions, managed policies, testing in non-prod, gradual rollout*
8. **Design an IAM strategy for a multi-tenant SaaS application with customer data isolation.**
    - *Hint: Dynamic policies, attribute-based access control, resource tagging, tenant isolation*

Identity Center (AWS SSO)

1. **How does AWS Identity Center integrate with external identity providers?**
    - *Hint: SAML 2.0, SCIM for user provisioning, Active Directory integration*
2. **What are permission sets in Identity Center?**
    - *Hint: Collections of policies assigned to users/groups for specific AWS accounts*
3. **Design a complex permission set strategy for a large enterprise with multiple business units and environments.**
    - *Hint: Hierarchical permission sets, session duration, inline vs managed policies, account assignment*
4. **How would you implement just-in-time access using Identity Center for privileged operations?**
    - *Hint: Time-bound access, approval workflows, temporary permission elevation, audit trails*
5. **Explain the integration between Identity Center and external SCIM providers for automated user lifecycle management.**
    - *Hint: SCIM protocol, user provisioning/deprovisioning, group membership sync, attribute mapping*
6. **How do you troubleshoot SAML authentication failures in Identity Center?**
    - *Hint: SAML response analysis, attribute mapping, clock skew, certificate validation*
7. **Design a solution for Identity Center to work with multiple Active Directory forests.**
    - *Hint: AD Connector, trust relationships, forest-to-forest trusts, user principal name suffixes*

S3 (Simple Storage Service)

1. **Explain S3 storage classes and their use cases.**
    - *Hint: Standard, IA, One Zone-IA, Glacier, Glacier Deep Archive, Intelligent Tiering*

2. **What is S3 Cross-Region Replication and its requirements?**
    - *Hint: Versioning enabled, different regions, IAM role for replication*
3. **How does S3 Transfer Acceleration work?**
    - *Hint: CloudFront edge locations, optimized network paths*
4. **Design a complex S3 lifecycle policy for a data lake with multiple data tiers and compliance requirements.**
    - *Hint: Transition rules, expiration rules, incomplete multipart uploads, object tagging, legal hold*
5. **How would you implement S3 event-driven processing for millions of objects with guaranteed delivery?**
    - *Hint: S3 Event Notifications + SQS + DLQ + Lambda + batch processing + idempotency*
6. **Explain S3 request patterns and how to optimize for high-throughput workloads.**
    - *Hint: Request rate performance, hotspotting, prefix distribution, multipart uploads, transfer acceleration*
7. **Design a solution for S3 data encryption with customer-managed keys and key rotation.**
    - *Hint: SSE-KMS, customer-managed CMKs, key rotation policies, cross-region key access*
8. **How do you implement S3 access logging and monitoring for compliance and security analysis?**
    - *Hint: Server access logs, CloudTrail data events, VPC Flow Logs, GuardDuty, access patterns analysis*
9. **Design a multi-region S3 disaster recovery strategy with RTO < 15 minutes.**
    - *Hint: Cross-Region Replication, S3 Transfer Acceleration, Route 53 failover, automated failover testing*

RDS (Relational Database Service)

1. **What are RDS Multi-AZ deployments vs Read Replicas?**
    - *Hint: Multi-AZ = HA/failover, Read Replicas = performance/scaling*
2. **Explain RDS backup strategies.**
    - *Hint: Automated backups, manual snapshots, point-in-time recovery*
3. **Design a complex RDS architecture for a global application with read/write splitting and disaster recovery.**
    - *Hint: Multi-region setup, read replicas, Aurora Global Database, connection pooling, failover automation*
4. **How would you migrate a 50TB on-premises Oracle database to RDS with minimal downtime?**
    - *Hint: DMS with CDC, SCT for schema conversion, parallel processing, validation strategies*
5. **Explain RDS Proxy and its benefits for serverless applications.**
    - *Hint: Connection pooling, IAM authentication, failover handling, Lambda integration*

6. **How do you implement RDS encryption at rest and in transit for existing databases?**
    - *Hint: Snapshot encryption, SSL/TLS certificates, parameter groups, application changes*
7. **Design a solution for RDS performance monitoring and automated scaling.**
    - *Hint: CloudWatch metrics, Performance Insights, automated storage scaling, read replica scaling*
8. **How would you implement cross-region RDS disaster recovery with automated failover?**
    - *Hint: Cross-region read replicas, Route 53 health checks, Lambda automation, data consistency*

VPC (Virtual Private Cloud)

1. **What is the difference between Security Groups and NACLs?**
    - *Hint: SG = stateful/instance level, NACL = stateless/subnet level*
2. **Explain VPC peering limitations.**
    - *Hint: No transitive peering, no overlapping CIDR blocks*
3. **What are VPC endpoints and their types?**
    - *Hint: Gateway endpoints (S3, DynamoDB), Interface endpoints (PrivateLink)*
4. **Design a complex multi-tier VPC architecture with DMZ, application, and database tiers across 3 AZs.**
    - *Hint: Public/private subnets, NAT gateways, route tables, security groups, NACLs, bastion hosts*
5. **How would you implement VPC Flow Logs analysis for security monitoring and network troubleshooting?**
    - *Hint: Flow Logs to CloudWatch/S3, Athena queries, GuardDuty integration, custom analysis*
6. **Explain Transit Gateway and design a hub-and-spoke network architecture for 50+ VPCs.**
    - *Hint: Transit Gateway, route tables, propagation, association, cross-region peering*
7. **How do you implement network segmentation using VPC and security groups for PCI compliance?**
    - *Hint: Network isolation, least privilege, monitoring, encryption, audit trails*
8. **Design a solution for on-premises to AWS connectivity with redundancy and high bandwidth.**
    - *Hint: Direct Connect, VPN backup, BGP routing, LAG, virtual interfaces*
9. **How would you troubleshoot intermittent connectivity issues in a complex VPC setup?**
    - *Hint: VPC Flow Logs, route table analysis, security group rules, DNS resolution, packet capture*

Lambda

1. **What are Lambda cold starts and how to minimize them?**
   - *Hint: Provisioned concurrency, connection pooling, smaller deployment packages*
2. **Explain Lambda execution context reuse.**
   - *Hint: Code outside handler function, global variables, database connections*
3. **Design a Lambda-based solution that processes 1 million events per minute with guaranteed ordering.**
   - *Hint: Kinesis Data Streams, Lambda concurrency, DLQ, batch processing, checkpointing*
4. **How would you implement Lambda function versioning and blue-green deployments?**
   - *Hint: Aliases, weighted routing, CodeDeploy, gradual rollout, rollback strategies*
5. **Explain Lambda layers and design a strategy for shared dependencies across multiple functions.**
   - *Hint: Layer versioning, dependency management, size limits, runtime compatibility*
6. **How do you implement Lambda function monitoring and observability at scale?**
   - *Hint: X-Ray tracing, custom metrics, structured logging, distributed tracing, alerting*
7. **Design a serverless architecture using Lambda for real-time data processing with exactly-once semantics.**
   - *Hint: Kinesis, DynamoDB, idempotency keys, error handling, retry logic*
8. **How would you optimize Lambda function performance for CPU-intensive workloads?**
   - *Hint: Memory allocation, multi-threading, connection pooling, warm-up strategies*

Step Functions

1. **What are the different Step Function workflow types?**
   - *Hint: Standard (exactly-once, audit), Express (high-volume, at-least-once)*
2. **Explain Step Function error handling patterns.**
   - *Hint: Retry, Catch, Fail states, exponential backoff*
3. **Design a complex Step Functions workflow for a multi-step data processing pipeline with conditional branching.**
   - *Hint: Choice states, parallel execution, Map state, error handling, state machine nesting*
4. **How would you implement long-running workflows with human approval steps using Step Functions?**
   - *Hint: Wait states, activity tasks, callback patterns, timeout handling, notification integration*
5. **Explain Step Functions integration patterns and when to use each.**
   - *Hint: Request-response, run job, wait for callback, optimized integrations*
6. **Design a Step Functions solution for orchestrating microservices with saga pattern implementation.**
   - *Hint: Compensating transactions, rollback logic, distributed state management*

7. **How do you implement Step Functions monitoring and debugging for complex workflows?**
    - *Hint: CloudWatch integration, X-Ray tracing, execution history, visual workflow debugging*

Load Balancers (ALB/NLB)

1. **When would you use ALB vs NLB vs CLB?**
    - *Hint: ALB = Layer 7/HTTP, NLB = Layer 4/TCP/extreme performance, CLB = legacy*
2. **What are ALB target groups and routing rules?**
    - *Hint: Path-based, host-based, header-based routing*
3. **Design a complex ALB configuration for a microservices architecture with canary deployments.**
    - *Hint: Weighted target groups, listener rules, health checks, blue-green deployment*
4. **How would you implement SSL termination and end-to-end encryption with ALB?**
    - *Hint: SSL certificates, SNI, backend encryption, certificate management*
5. **Explain NLB cross-zone load balancing and its impact on availability and performance.**
    - *Hint: Traffic distribution, AZ failure scenarios, latency considerations, cost implications*
6. **How do you implement sticky sessions with ALB for stateful applications?**
    - *Hint: Application-based cookies, duration-based cookies, session affinity limitations*
7. **Design a solution for global load balancing using multiple ALBs across regions.**
    - *Hint: Route 53 latency-based routing, health checks, failover scenarios*
8. **How would you troubleshoot ALB 5xx errors and high latency issues?**
    - *Hint: Access logs, CloudWatch metrics, target health, connection draining*

ECS (Elastic Container Service)

1. **What is the difference between ECS EC2 launch type vs Fargate?**
    - *Hint: EC2 = manage infrastructure, Fargate = serverless containers*
2. **Explain ECS task definitions and services.**
    - *Hint: Task definition = blueprint, Service = desired count management*
3. **Design a complex ECS architecture for a microservices application with service discovery and load balancing.**
    - *Hint: Service Connect, Cloud Map, ALB integration, task networking, security groups*
4. **How would you implement ECS auto-scaling based on custom metrics?**
    - *Hint: Application Auto Scaling, custom CloudWatch metrics, target tracking, step scaling*
5. **Explain ECS task placement strategies and constraints.**

   - *Hint: Binpack, random, spread strategies, attribute-based constraints, AZ distribution*
6. **How do you implement blue-green deployments for ECS services?**
   - *Hint: CodeDeploy integration, target group swapping, rollback mechanisms*
7. **Design an ECS solution for batch processing with spot instances and fault tolerance.**
   - *Hint: Spot fleet, mixed instance types, checkpointing, queue-based processing*
8. **How would you implement ECS logging and monitoring for containerized applications?**
   - *Hint: CloudWatch Logs, log drivers, container insights, X-Ray integration*

EKS (Elastic Kubernetes Service)

1. **What are EKS node groups and their types?**
   - *Hint: Managed node groups, self-managed nodes, Fargate profiles*
2. **How does EKS integrate with AWS services?**
   - *Hint: ALB Ingress Controller, EBS CSI driver, IAM roles for service accounts*
3. **Design a complex EKS cluster architecture with multiple node groups and mixed instance types.**
   - *Hint: Spot instances, on-demand instances, node selectors, taints and tolerations*
4. **How would you implement EKS cluster autoscaling and pod autoscaling?**
   - *Hint: Cluster Autoscaler, HPA, VPA, KEDA, custom metrics*
5. **Explain EKS networking with VPC CNI and design for high-performance workloads.**
   - *Hint: ENI allocation, IP address management, network policies, bandwidth optimization*
6. **How do you implement EKS security best practices including RBAC and pod security?**
   - *Hint: IAM roles for service accounts, RBAC, pod security standards, network policies*
7. **Design an EKS GitOps workflow with automated deployments and rollbacks.**
   - *Hint: ArgoCD, Flux, Git repositories, automated testing, progressive delivery*
8. **How would you implement EKS disaster recovery across multiple regions?**
   - *Hint: Cross-region replication, backup strategies, stateful workload migration*
9. **Explain EKS add-ons and how to manage them at scale.**
   - *Hint: Managed add-ons, Helm charts, operator patterns, lifecycle management*

CloudWatch

1. **What are CloudWatch custom metrics and how to create them?**
   - *Hint: PutMetricData API, CloudWatch agent, custom namespaces*
2. **Explain CloudWatch Logs Insights query language.**
   - *Hint: SQL-like syntax, fields, filter, stats commands*
3. **Design a comprehensive CloudWatch monitoring strategy for a multi-tier application.**
   - *Hint: Custom metrics, composite alarms, dashboards, anomaly detection, cross-service correlation*

4. **How would you implement CloudWatch Logs aggregation and analysis for 1000+ microservices?**
    - *Hint: Log groups, metric filters, Kinesis Data Firehose, centralized logging, cost optimization*
5. **Explain CloudWatch Events/EventBridge patterns and design an event-driven architecture.**
    - *Hint: Event patterns, rules, targets, custom event buses, cross-account events*
6. **How do you implement CloudWatch Container Insights for EKS clusters at scale?**
    - *Hint: FluentD/Fluent Bit, performance monitoring, resource utilization, cost analysis*
7. **Design a CloudWatch alerting strategy with escalation and automated remediation.**
    - *Hint: Composite alarms, SNS topics, Lambda automation, PagerDuty integration*

CloudTrail

1. **What is the difference between CloudTrail data events vs management events?**
    - *Hint: Management = control plane, Data = data plane (S3 object access)*
2. **How to set up CloudTrail for multi-account logging?**
    - *Hint: Organization trail, cross-account S3 bucket policies*
3. **Design a CloudTrail architecture for compliance and security monitoring across 100+ AWS accounts.**
    - *Hint: Organization trails, log file validation, encryption, centralized analysis*
4. **How would you implement real-time CloudTrail log analysis for security incident response?**
    - *Hint: CloudWatch Logs, Kinesis, Lambda, EventBridge, automated response*
5. **Explain CloudTrail Insights and how to detect unusual API activity patterns.**
    - *Hint: Baseline establishment, anomaly detection, write events analysis*
6. **How do you implement CloudTrail log integrity verification and tamper detection?**
    - *Hint: Log file validation, digest files, cryptographic verification*
7. **Design a solution for CloudTrail cost optimization while maintaining security coverage.**
    - *Hint: Event filtering, S3 lifecycle policies, data events selection, log analysis optimization*

Config

1. **What are AWS Config rules and remediation actions?**
    - *Hint: Compliance rules, automatic remediation with Systems Manager*
2. **How does Config track resource relationships?**
    - *Hint: Configuration items, relationship mapping, timeline view*
3. **Design a comprehensive AWS Config strategy for enterprise compliance management.**
    - *Hint: Organization Config, aggregators, conformance packs, multi-account rules*
4. **How would you implement custom Config rules for organization-specific compliance requirements?**

    - *Hint: Lambda-based rules, periodic vs configuration change triggers, rule parameters*
5. **Explain Config remediation and design automated compliance enforcement.**
    - *Hint: Systems Manager automation, manual remediation, resource relationships*
6. **How do you implement Config cost optimization while maintaining compliance coverage?**
    - *Hint: Resource type selection, delivery channel optimization, rule targeting*
7. **Design a Config-based solution for tracking resource changes and impact analysis.**
    - *Hint: Configuration history, relationship tracking, change notifications, timeline analysis*

Security Hub

1. **How does Security Hub aggregate findings from multiple services?**
    - *Hint: AWS Security Finding Format (ASFF), integrations with GuardDuty, Inspector*
2. **What are Security Hub compliance standards?**
    - *Hint: AWS Foundational Security Standard, CIS, PCI DSS*
3. **Design a Security Hub architecture for centralized security monitoring across multiple AWS accounts.**
    - *Hint: Master-member accounts, cross-region aggregation, custom insights, automated response*
4. **How would you implement custom Security Hub findings from third-party security tools?**
    - *Hint: ASFF format, BatchImportFindings API, finding normalization, severity mapping*
5. **Explain Security Hub custom insights and design security metrics dashboards.**
    - *Hint: Finding filters, grouping criteria, trend analysis, executive reporting*
6. **How do you implement Security Hub automated remediation workflows?**
    - *Hint: EventBridge integration, Lambda functions, Systems Manager, playbook automation*
7. **Design a Security Hub solution for compliance reporting and audit preparation.**
    - *Hint: Compliance scores, evidence collection, automated reporting, historical tracking*

Control Tower

1. **What are Control Tower guardrails?**
    - *Hint: Preventive (SCPs), Detective (Config rules), mandatory vs optional*
2. **Explain Control Tower Account Factory.**
    - *Hint: Automated account provisioning, baseline configurations*
3. **Design a Control Tower landing zone for a large enterprise with complex organizational requirements.**

- *Hint: OU structure, custom guardrails, account factory customization, governance at scale*

4. **How would you implement custom guardrails in Control Tower for organization-specific policies?**
    - *Hint: Custom SCPs, Config rules, Lambda-based controls, lifecycle management*
5. **Explain Control Tower drift detection and remediation strategies.**
    - *Hint: Drift detection, manual changes, automated remediation, governance enforcement*
6. **How do you extend Control Tower with additional AWS services and third-party integrations?**
    - *Hint: Customizations for Control Tower, CloudFormation, lifecycle events*
7. **Design a Control Tower migration strategy for existing multi-account environments.**
    - *Hint: Account enrollment, existing resource handling, minimal disruption, rollback planning*

SCPs (Service Control Policies)

1. **How do SCPs work in AWS Organizations?**
    - *Hint: Permission boundaries, inheritance, explicit deny only*
2. **What are common SCP use cases?**
    - *Hint: Restrict regions, prevent root user actions, enforce tagging*
3. **Design complex SCPs for a multi-business unit organization with different compliance requirements.**
    - *Hint: OU-specific policies, condition-based restrictions, exception handling, policy inheritance*
4. **How would you implement SCPs for data residency and sovereignty requirements?**
    - *Hint: Region restrictions, service limitations, cross-border data transfer controls*
5. **Explain SCP testing and validation strategies before production deployment.**
    - *Hint: Policy simulator, test accounts, gradual rollout, impact analysis*
6. **How do you implement SCPs for cost control and resource governance?**
    - *Hint: Instance type restrictions, spending limits, resource tagging enforcement*
7. **Design SCPs for security compliance including encryption and access controls.**
    - *Hint: Encryption enforcement, public access prevention, privileged action restrictions*

WAF (Web Application Firewall)

1. **What are WAF rule types and actions?**
    - *Hint: Allow, Block, Count actions; Rate-based, IP, String match rules*
2. **How does WAF integrate with CloudFront and ALB?**
    - *Hint: Web ACLs association, real-time metrics*
3. **Design a comprehensive WAF strategy for protecting a multi-tier web application.**

- *Hint: Managed rule groups, custom rules, rate limiting, geo-blocking, bot protection*

4. **How would you implement WAF with machine learning for advanced threat detection?**
    - *Hint: Intelligent threat mitigation, anomaly detection, adaptive rules, false positive reduction*
5. **Explain WAF logging and monitoring for security analysis and tuning.**
    - *Hint: Web ACL logs, CloudWatch metrics, Kinesis integration, log analysis*
6. **How do you implement WAF rule testing and deployment automation?**
    - *Hint: Count mode testing, A/B testing, automated rule updates, performance impact*
7. **Design a WAF solution for API protection with rate limiting and authentication.**
    - *Hint: API-specific rules, token validation, request signing, DDoS protection*

Shield

1. **What is the difference between Shield Standard and Advanced?**
    - *Hint: Standard = free DDoS protection, Advanced = enhanced protection + DRT support*
2. **Design a comprehensive DDoS protection strategy using Shield Advanced.**
    - *Hint: Global accelerator, Route 53, CloudFront, cost protection, DRT engagement*
3. **How would you implement Shield Advanced monitoring and alerting?**
    - *Hint: CloudWatch metrics, SNS notifications, attack visibility, response automation*
4. **Explain Shield Advanced cost protection and how to optimize DDoS-related charges.**
    - *Hint: Cost protection policies, usage spike protection, billing analysis*
5. **How do you integrate Shield with WAF for layered DDoS protection?**
    - *Hint: Combined protection strategies, rule coordination, attack mitigation*

Inspector

1. **What types of assessments does Inspector perform?**
    - *Hint: Network reachability, security vulnerabilities, runtime behavior*
2. **Design an Inspector strategy for continuous security assessment of EC2 and container workloads.**
    - *Hint: Assessment targets, rules packages, automated scanning, finding prioritization*
3. **How would you implement Inspector findings integration with CI/CD pipelines?**
    - *Hint: API integration, automated blocking, vulnerability gates, remediation workflows*
4. **Explain Inspector network reachability analysis and security group optimization.**
    - *Hint: Network path analysis, port accessibility, security group recommendations*
5. **How do you implement Inspector at scale for large container environments?**
    - *Hint: ECR integration, automated scanning, policy enforcement, exception handling*

GuardDuty

1. **What data sources does GuardDuty analyze?**
   - *Hint: VPC Flow Logs, DNS logs, CloudTrail events, threat intelligence*
2. **How to respond to GuardDuty findings?**
   - *Hint: EventBridge integration, automated remediation, custom actions*
3. **Design a GuardDuty architecture for multi-account threat detection and response.**
   - *Hint: Master-member accounts, centralized findings, automated response, threat hunting*
4. **How would you implement GuardDuty custom threat intelligence and IOCs?**
   - *Hint: Threat intel feeds, custom IP lists, domain lists, automated updates*
5. **Explain GuardDuty machine learning capabilities and anomaly detection.**
   - *Hint: Behavioral analysis, baseline establishment, anomaly scoring, false positive reduction*
6. **How do you implement GuardDuty findings correlation with SIEM systems?**
   - *Hint: Finding export, SIEM integration, correlation rules, incident enrichment*
7. **Design GuardDuty automated remediation for different threat types.**
   - *Hint: Lambda functions, Systems Manager, network isolation, credential rotation*

Fargate

1. **What are Fargate task sizing considerations?**
   - *Hint: CPU/memory combinations, pricing model, resource allocation*
2. **Design a Fargate architecture for microservices with auto-scaling and cost optimization.**
   - *Hint: Task sizing, scaling policies, Spot Fargate, resource utilization*
3. **How would you implement Fargate networking and security best practices?**
   - *Hint: VPC configuration, security groups, task roles, secrets management*
4. **Explain Fargate monitoring and observability strategies.**
   - *Hint: Container Insights, custom metrics, distributed tracing, log aggregation*
5. **How do you implement Fargate CI/CD with blue-green deployments?**
   - *Hint: CodeDeploy integration, traffic shifting, rollback strategies, health checks*

CodeDeploy

1. **What are CodeDeploy deployment configurations?**
   - *Hint: Blue/Green, Rolling, All-at-once, custom configurations*
2. **Design a CodeDeploy strategy for zero-downtime deployments across multiple environments.**
   - *Hint: Blue-green deployments, canary releases, automated rollbacks, health checks*
3. **How would you implement CodeDeploy with Lambda for serverless application deployments?**

- *Hint: Alias shifting, traffic splitting, automated rollback, monitoring integration*

4. **Explain CodeDeploy hooks and custom deployment scripts.**
   - *Hint: Lifecycle events, custom scripts, validation steps, cleanup procedures*
5. **How do you implement CodeDeploy monitoring and automated rollback triggers?**
   - *Hint: CloudWatch alarms, custom metrics, rollback automation, notification systems*

CodePipeline

1. **How to implement approval gates in CodePipeline?**
   - *Hint: Manual approval actions, SNS notifications, IAM permissions*
2. **Design a complex CodePipeline for multi-environment deployments with parallel stages.**
   - *Hint: Parallel actions, environment promotion, approval gates, artifact management*
3. **How would you implement CodePipeline with cross-account deployments?**
   - *Hint: Cross-account roles, artifact buckets, KMS encryption, permission boundaries*
4. **Explain CodePipeline integration with third-party tools and custom actions.**
   - *Hint: Custom action types, webhook triggers, external integrations, API calls*
5. **How do you implement CodePipeline monitoring and failure handling?**
   - *Hint: CloudWatch events, failure notifications, retry mechanisms, pipeline optimization*

CodeBuild

1. **What are CodeBuild build environments and specifications?**
   - *Hint: buildspec.yml, environment variables, artifacts*
2. **Design a CodeBuild strategy for multi-language, multi-environment builds.**
   - *Hint: Custom build images, build matrix, parallel builds, artifact optimization*
3. **How would you implement CodeBuild with Docker and container image builds?**
   - *Hint: Docker in Docker, ECR integration, image scanning, multi-stage builds*
4. **Explain CodeBuild caching strategies for build optimization.**
   - *Hint: S3 caching, local caching, dependency caching, build time optimization*
5. **How do you implement CodeBuild security scanning and compliance checks?**
   - *Hint: SAST tools, dependency scanning, compliance validation, quality gates*

Terraform

1. **How to manage Terraform state in AWS?**
   - *Hint: S3 backend, DynamoDB locking, state encryption*
2. **Design a Terraform architecture for multi-environment, multi-account AWS deployments.**

    - *Hint: Workspace management, remote state, module organization, cross-account roles*
3. **How would you implement Terraform CI/CD with automated testing and validation?**
    - *Hint: Plan validation, automated testing, drift detection, approval workflows*
4. **Explain Terraform module design patterns and best practices for reusability.**
    - *Hint: Module composition, versioning, input validation, output design*
5. **How do you implement Terraform state management and disaster recovery?**
    - *Hint: State backup, versioning, recovery procedures, state migration*
6. **Design Terraform security practices including secrets management.**
    - *Hint: Variable encryption, provider authentication, state security, policy as code*

CloudFormation

1. **What are CloudFormation stack policies and drift detection?**
    - *Hint: Prevent updates, detect manual changes, remediation*
2. **Design a CloudFormation architecture for large-scale, multi-account deployments.**
    - *Hint: Nested stacks, cross-stack references, StackSets, parameter management*
3. **How would you implement CloudFormation custom resources for complex requirements?**
    - *Hint: Lambda-backed resources, provider framework, lifecycle management*
4. **Explain CloudFormation rollback strategies and failure handling.**
    - *Hint: Rollback triggers, stack policies, deletion policies, failure modes*
5. **How do you implement CloudFormation template validation and testing?**
    - *Hint: Template linting, parameter validation, integration testing, drift detection*
6. **Design CloudFormation CI/CD with automated deployments and approvals.**
    - *Hint: Pipeline integration, change sets, approval processes, automated testing*

Easy Questions

1. What is AWS and what are its main benefits?

Hint: Cloud computing platform, pay-as-you-go, global infrastructure, scalability

2. What are AWS Regions and Availability Zones?

Hint: Regions = geographic areas, AZs = isolated data centers within regions

3. What is the difference between horizontal and vertical scaling?

Hint: Horizontal = add more instances, Vertical = increase instance size

4. What is an AMI in EC2?

Hint: Amazon Machine Image, template for launching instances

5. What are the main S3 storage classes?

Hint: Standard, Standard-IA, One Zone-IA, Glacier, Glacier Deep Archive

6. What is the purpose of IAM in AWS?

Hint: Identity and Access Management, control who can access what

7. What is a VPC?

Hint: Virtual Private Cloud, isolated network environment

8. What is the difference between public and private subnets?

Hint: Public = internet gateway route, Private = no direct internet access

9. What is CloudWatch used for?

Hint: Monitoring, logging, metrics, alarms

10. What is the difference between EBS and Instance Store?

Hint: EBS = persistent network storage, Instance Store = temporary local storage

11. What is Auto Scaling?

Hint: Automatically adjust number of instances based on demand

12. What is CloudFront?

Hint: Content Delivery Network (CDN), global edge locations

13. What is Route 53?

Hint: DNS service, domain registration, health checks

14. What is the difference between RDS and DynamoDB?

Hint: RDS = relational database, DynamoDB = NoSQL database

15. What is Lambda?

Hint: Serverless compute service, event-driven execution

16. What is the AWS Shared Responsibility Model?

Hint: AWS = security OF the cloud, Customer = security IN the cloud

17. What is CloudTrail?

Hint: API logging service, audit trail of AWS account activity

18. What is the difference between Security Groups and NACLs?

Hint: Security Groups = instance-level firewall, NACLs = subnet-level firewall

19. What is an Elastic IP?

Hint: Static public IP address that can be attached to instances

20. What is SNS?

Hint: Simple Notification Service, pub/sub messaging

Scenario-Based Questions (25)

1. Your web application experiences traffic spikes during business hours. How would you design an auto-scaling solution?

Hint: ALB + Auto Scaling Groups + CloudWatch metrics + scaling policies + warm-up periods

2. You need to migrate a 10TB database to AWS with minimal downtime. What approach would you take?

Hint: DMS with CDC, read replicas, blue-green deployment, Route 53 weighted routing

3. Your company requires all data to be encrypted at rest and in transit. How would you implement this across S3, RDS, and EBS?

Hint: S3 SSE, RDS encryption, EBS encryption, KMS keys, SSL/TLS certificates

4. Design a disaster recovery solution for a critical application with RTO of 4 hours and RPO of 1 hour.

Hint: Multi-region deployment, automated backups, cross-region replication, Route 53 health checks

5. You need to provide secure access to AWS resources for a third-party vendor. How would you implement this?

Hint: Cross-account IAM roles, external ID, temporary credentials, least privilege

6. Your application needs to process files uploaded to S3 and store results in DynamoDB. Design the architecture.

Hint: S3 event notifications + Lambda + DynamoDB, error handling, DLQ

7. How would you implement a blue-green deployment for a containerized application?

Hint: ECS/EKS + ALB target groups + CodeDeploy + health checks

8. Design a solution to automatically remediate security violations detected by AWS Config.

Hint: Config rules + EventBridge + Lambda + Systems Manager automation

9. Your application requires consistent performance and cannot tolerate cold starts. How would you optimize Lambda?

Hint: Provisioned concurrency, connection pooling, smaller packages, warm-up strategies

10. Implement a solution to automatically scale DynamoDB based on traffic patterns.

Hint: DynamoDB Auto Scaling, CloudWatch metrics, target utilization

11. Design a multi-tier architecture with high availability and fault tolerance.

Hint: Multi-AZ deployment, ALB, Auto Scaling, RDS Multi-AZ, ElastiCache

12. How would you implement centralized logging for a microservices architecture?

Hint: CloudWatch Logs, Log Groups, Log Streams, Insights, centralized dashboard

13. Your organization needs to enforce compliance across multiple AWS accounts. Design the governance structure.

Hint: Organizations + Control Tower + SCPs + Config + Security Hub

14. Implement a solution to automatically backup and restore EC2 instances.

Hint: Data Lifecycle Manager, EBS snapshots, Lambda automation, cross-region copying

15. Design a cost-optimized solution for batch processing workloads.

Hint: Spot Instances, Batch service, SQS, Lambda, S3 lifecycle policies

16. How would you implement API rate limiting and throttling?

Hint: API Gateway throttling, WAF rate-based rules, Lambda concurrency limits

17. Design a solution to monitor and alert on application performance issues.

Hint: CloudWatch custom metrics, X-Ray tracing, alarms, SNS notifications

18. Implement a secure CI/CD pipeline with approval workflows.

Hint: CodeCommit + CodeBuild + CodeDeploy + CodePipeline + manual approvals + IAM roles

19. Your application needs to handle sudden traffic spikes of 10x normal load. Design the architecture.

Hint: CloudFront + ALB + Auto Scaling + SQS + Lambda + DynamoDB on-demand

20. How would you implement data archival with different retention policies?

Hint: S3 lifecycle policies, Glacier, legal hold, cross-region replication

21. Design a solution for real-time data processing and analytics.

Hint: Kinesis Data Streams + Kinesis Analytics + Lambda + DynamoDB + QuickSight

22. Implement a solution to automatically patch EC2 instances with zero downtime.

Hint: Systems Manager Patch Manager + Auto Scaling + rolling updates + health checks

23. How would you design a secure file sharing solution for external partners?

Hint: S3 pre-signed URLs, CloudFront signed URLs, time-limited access, encryption

24. Implement a solution to automatically scale based on custom business metrics.

Hint: Custom CloudWatch metrics + Auto Scaling policies + application-level monitoring

25. Design a multi-region active-active architecture for a global application.

Hint: Route 53 latency routing + multi-region deployment + DynamoDB Global Tables + cross-region replication

Complex Scenario-Based Questions (15)

1. Design a HIPAA-compliant healthcare application architecture with end-to-end encryption and audit trails.

Hint: VPC with private subnets + KMS + CloudTrail + Config + dedicated tenancy + BAA + encryption everywhere

2. Your organization needs to migrate 500+ applications to AWS with minimal business disruption. Design the migration strategy.

Hint: 6R strategy + Application Discovery Service + migration waves + automated testing + rollback procedures

3. Implement a zero-trust security model for a financial services application.

Hint: Identity Center + conditional access + device compliance + network segmentation + continuous monitoring

4. Design a solution to handle 1 million concurrent users with sub-100ms response times globally.

Hint: Global infrastructure + CloudFront + API Gateway + Lambda@Edge + DynamoDB Global Tables + caching strategies

5. Your company needs to implement data residency requirements across multiple countries while maintaining performance.

Hint: Multi-region architecture + data classification + regional data stores + cross-border data transfer controls

6. Design a solution to automatically detect and respond to security threats in real-time.

Hint: GuardDuty + Security Hub + EventBridge + Lambda + Systems Manager + automated remediation + SOAR integration

7. Implement a cost optimization strategy that reduces AWS spend by 40% without impacting performance.

Hint: Reserved Instances + Spot Instances + rightsizing + S3 lifecycle + unused resource identification + automation

8. Design a solution for processing 10TB of data daily with complex transformations and ML inference.

Hint: Kinesis + EMR + Glue + SageMaker + S3 + data lake architecture + parallel processing

9. Your application requires 99.99% uptime with automatic failover across regions in under 60 seconds.

Hint: Multi-region active-passive + Route 53 health checks + automated failover + data synchronization + testing procedures

10. Implement a solution to manage secrets and credentials across 1000+ applications with automatic rotation.

Hint: Secrets Manager + Parameter Store + Lambda rotation + IAM roles + encryption + audit logging

11. Design a serverless architecture that can handle unpredictable workloads ranging from 0 to 100,000 requests per second.

Hint: API Gateway + Lambda + DynamoDB on-demand + SQS + Step Functions + CloudFront + auto-scaling strategies

12. Your organization needs to implement a data mesh architecture with decentralized data ownership.

Hint: Lake Formation + Glue Data Catalog + cross-account access + data contracts + governance + self-service analytics

13. Design a solution for real-time fraud detection processing millions of transactions per minute.

Hint: Kinesis Data Streams + Kinesis Analytics + Lambda + DynamoDB + SageMaker + real-time ML inference

14. Implement a disaster recovery solution with automated testing and validation across multiple failure scenarios.

Hint: Multi-region replication + automated DR testing + chaos engineering + recovery procedures + compliance validation

15. Design a solution to migrate from on-premises Kubernetes to EKS while maintaining GitOps workflows and security policies.

Hint: EKS + Fargate + service mesh + GitOps tools + policy migration + security scanning + gradual migration